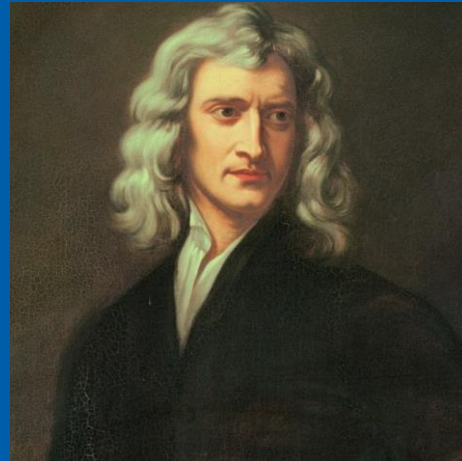


Astronomy 102 - Lab 6

Orbital Motion

Matthew Prest

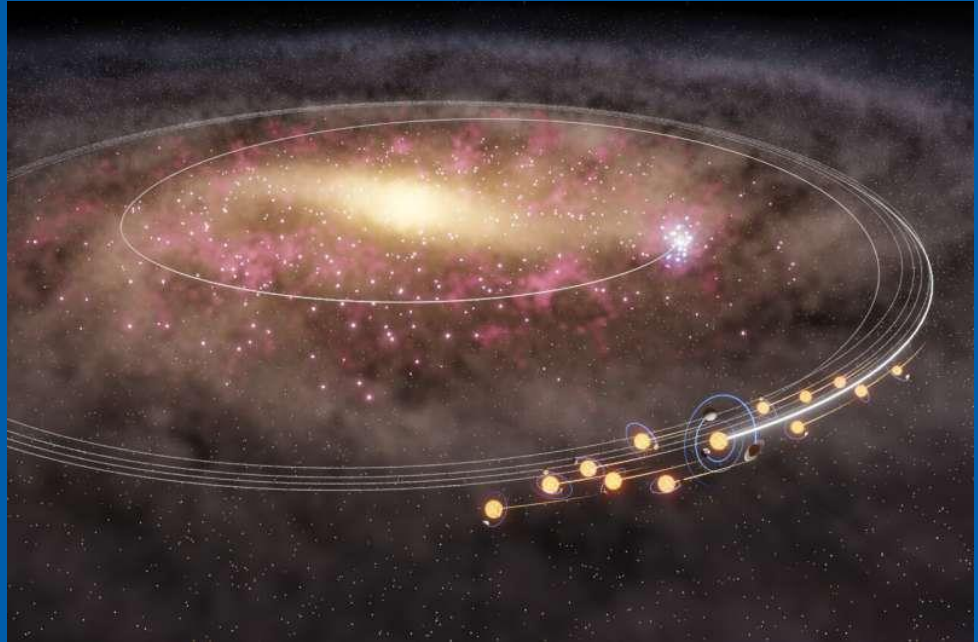


Copernicus: 1473 - 1543
Galileo: 1564 - 1642
Johannes Kepler: 1571 -1630
First Telescope: 1608
Newton: 1643 - 1727

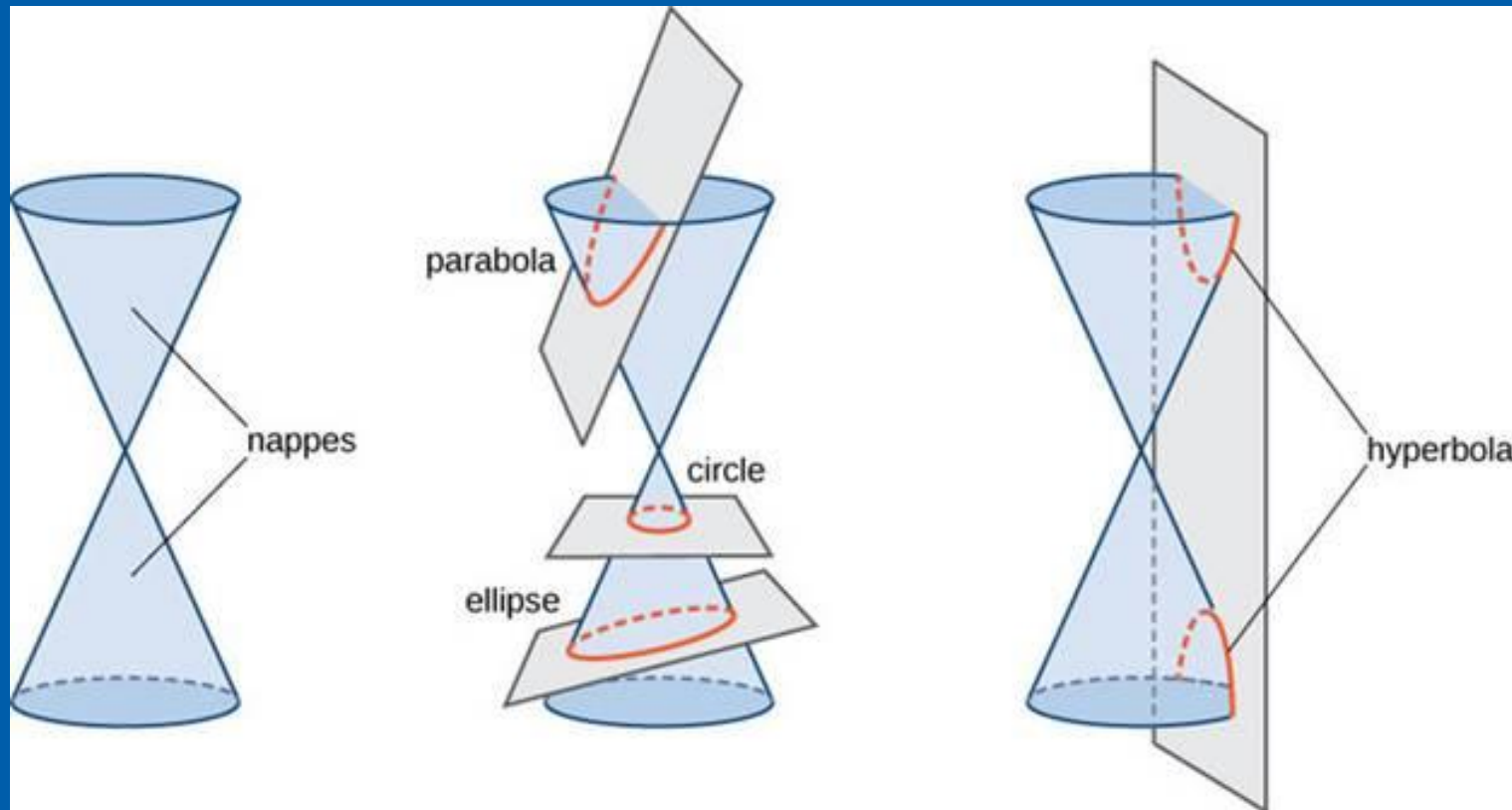
The latest in Astronomy

We are not alone:

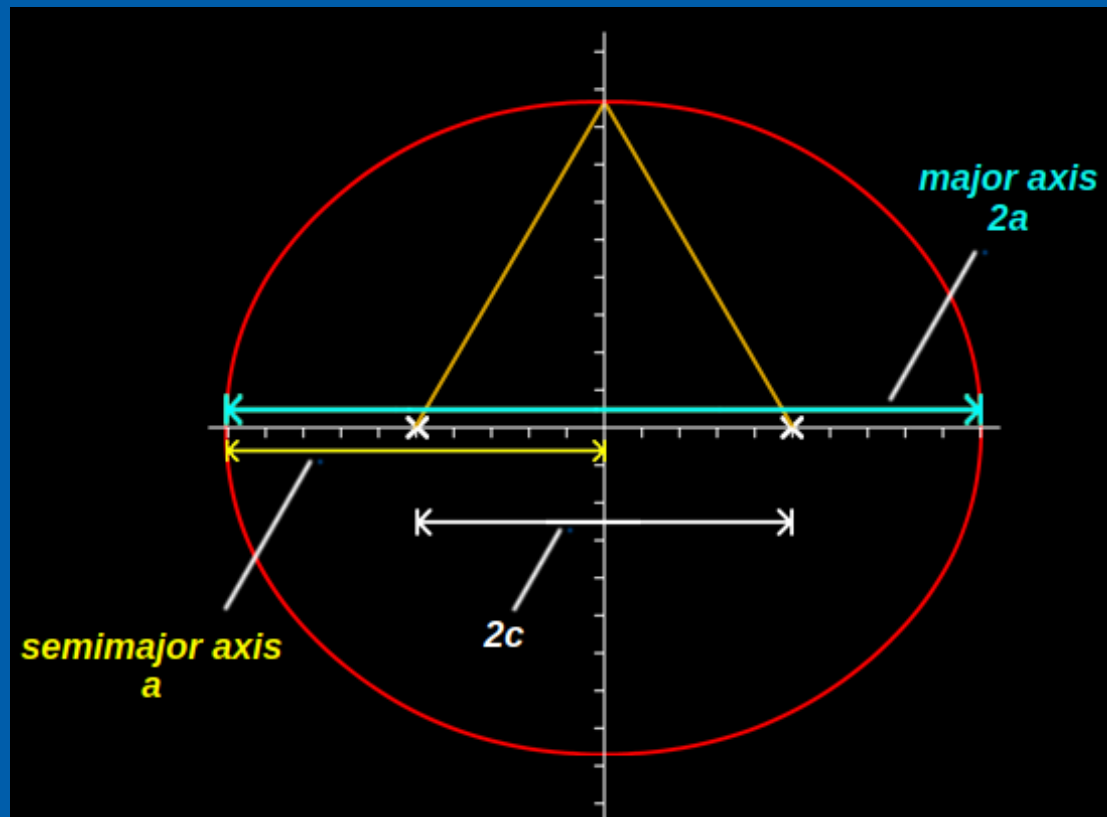
- Our Sun escaped together with stellar 'twins' from galaxy center
- Researchers have uncovered evidence for our sun joining a mass migration of similar "twins" leaving the core regions of our galaxy, 4 to 6 billion years ago



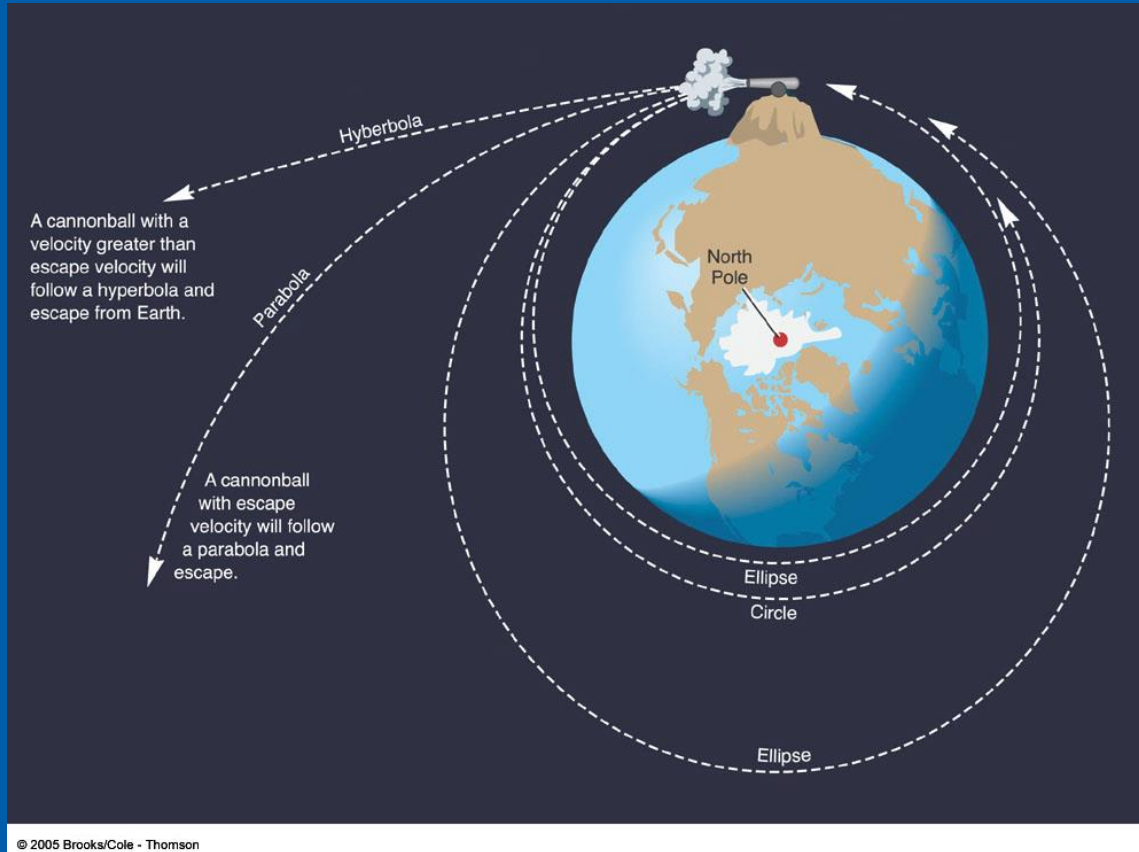
Conic Sections



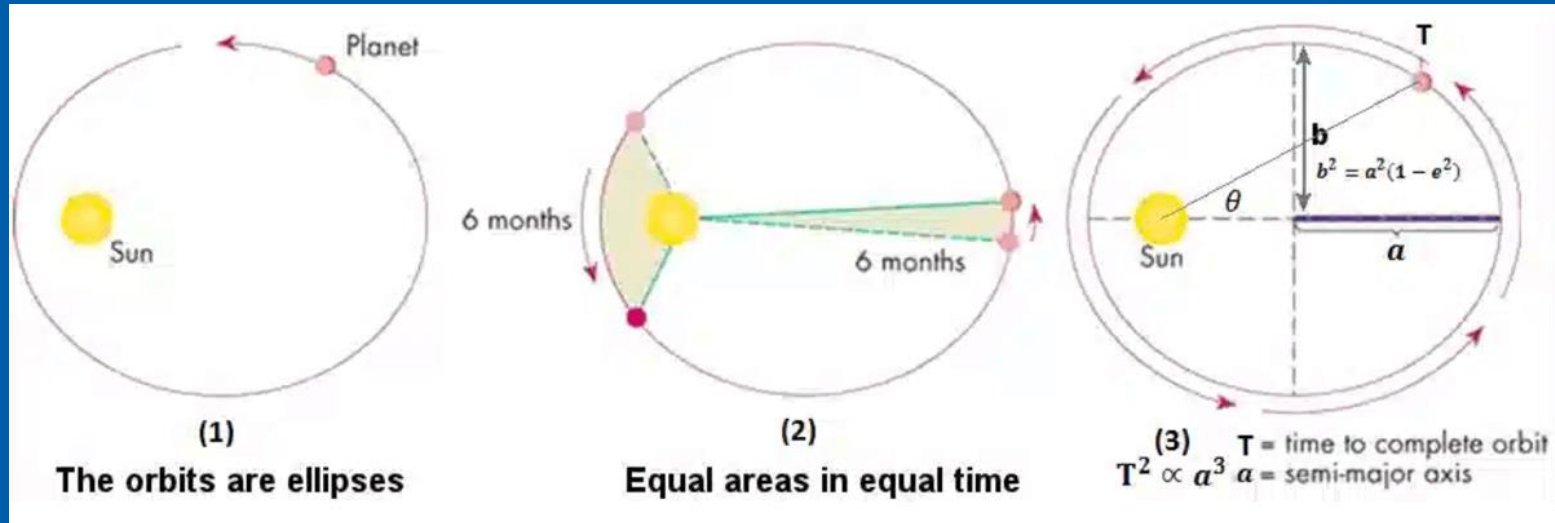
Conic Sections



Conic Sections

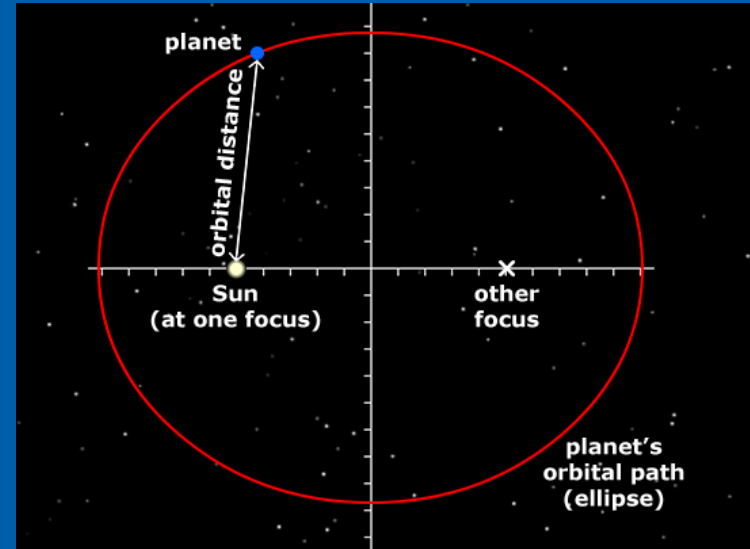
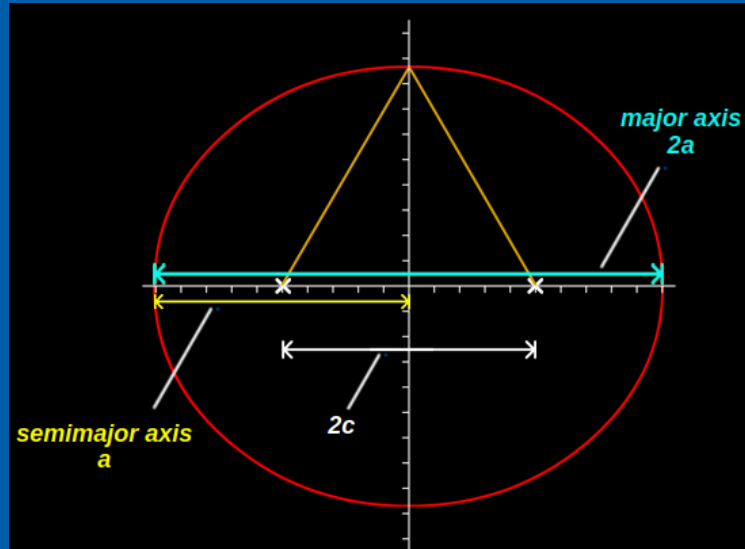


Kepler's 3 Laws of Planetary Motion

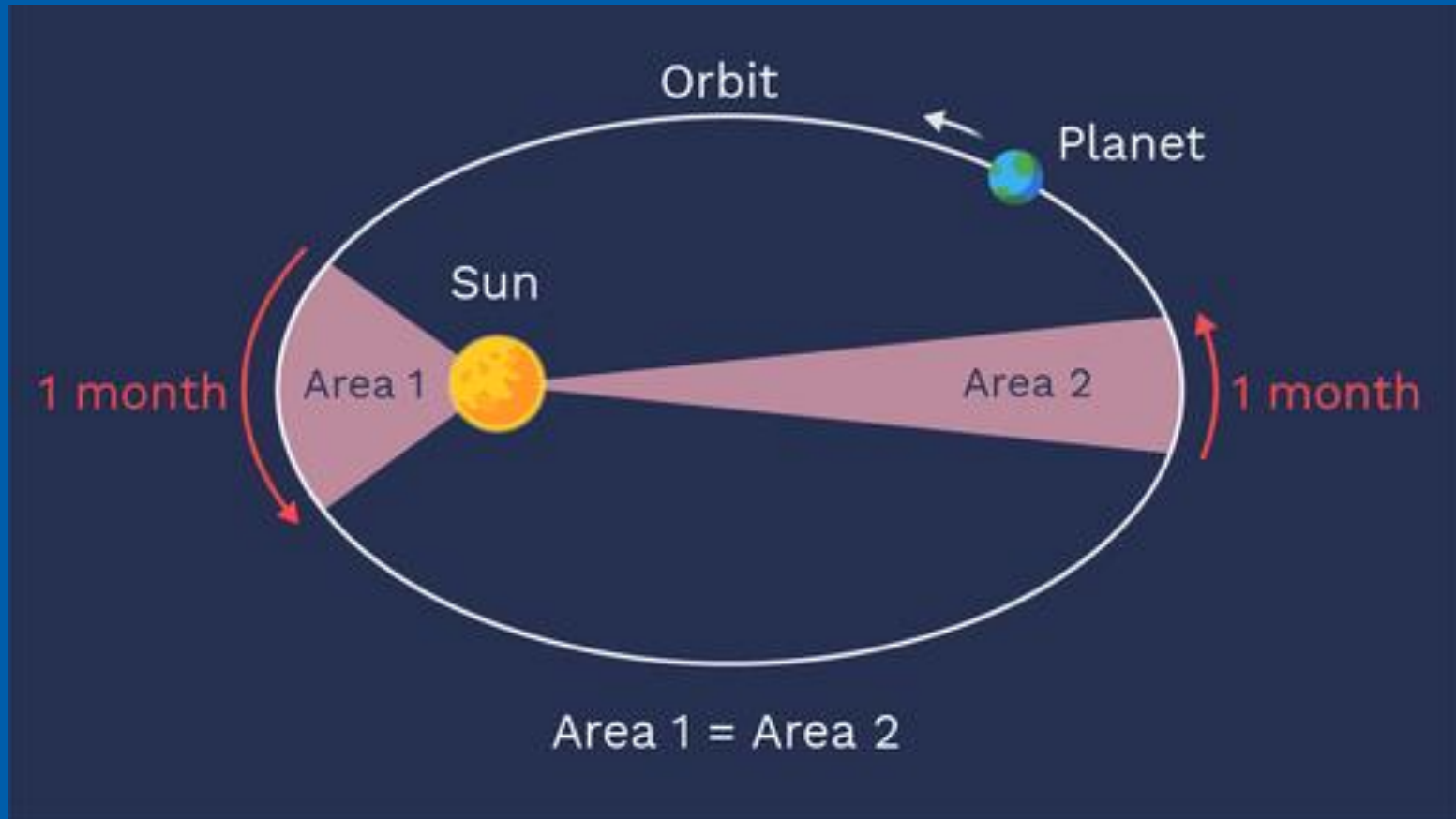


These laws are descriptions of motion, they do not describe causes.

First Law



Second Law



Third Law

- Kepler proposed the law purely in terms of proportionality
- With a little help from Newton we can figure out what that constant of proportionality is exactly!

$$T^2 = \frac{4\pi^2}{GM} a^3$$

can be expressed
as simply

$$T^2 = a^3$$

If expressed in the following units:

T Earth years

a Astronomical units AU
($a = 1$ AU for Earth)

M Solar masses $M_{\odot}$

Then $\frac{4\pi^2}{G} = 1$

Reminders

1. Complete the Tutorial first!

- This is best done now in class where I am available to assist

1. Attempt the essay questions last!

- These should be based on what you have covered in the tutorial.

1. Previous week's lab report due at the beginning of the next lab!

- Final grades are based on best 6 online labs plus each of the 4 in-class labs

1. Grades will be posted to Brightspace

- Check now that you have access to our course!
- Grade feedback will also be provided on Pearson, please check as this is the best way to improve your future grades!

1. Office hours

- Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!

Any and all questions are welcome!